

HAT-P-36b

R-0804

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250331 · created 2026-07-28T10:32:31+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460765.7268 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.128 +/- 0.011
Transit depth (Rp/Rs)^2	1.63 +/- 0.28 [%]
Orbital Inclination (inc)	87.0 +/- 2.12
Airmass coefficient 1 (a1)	1.019 +/- 0.011
Airmass coefficient 2 (a2)	-0.017 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	#1 - [100, 147]
Optimal Aperture	3.49
Optimal Annulus	12.19
Transit Duration (day)	0.0951 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.128 $\pm$ 0.011 vs 0.126 (deviation 0.14 σ)
Duration fit vs published	0.0951 d vs 0.0925 d (deviation 0.56 σ)
Mid-time O–C	0.0 min
Depth SNR	9.3
Residual scatter	1.11 %

Light curve

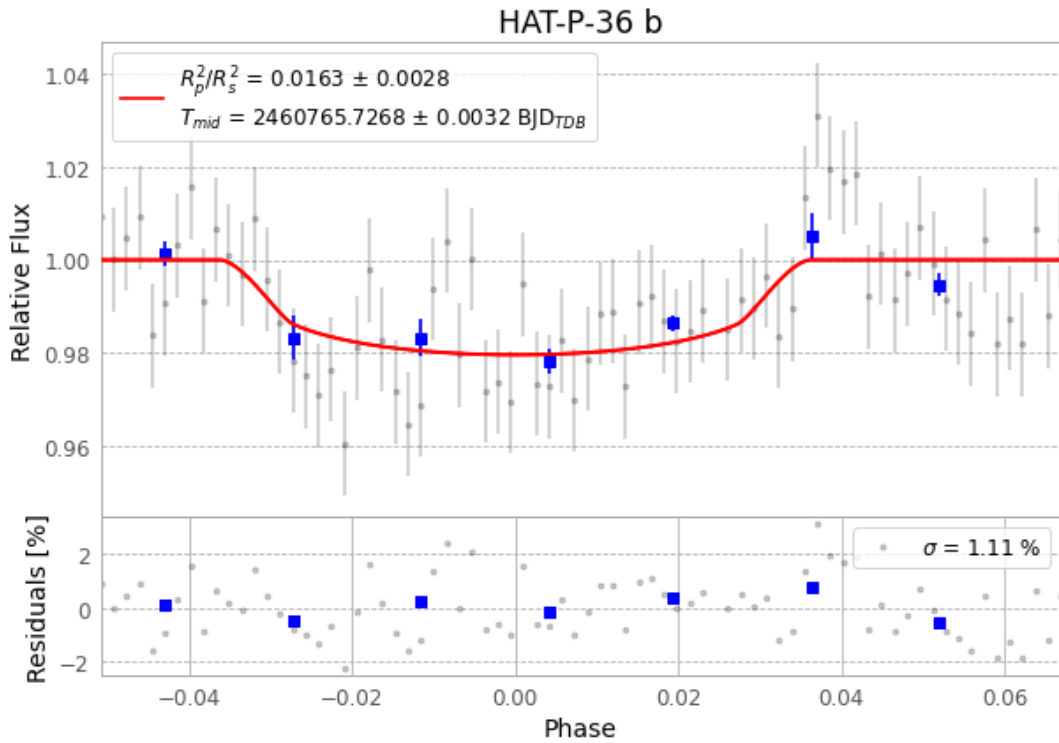


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-03-30.png (SHA-256 bd26e73d70a35856...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 214], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8c3e1680f6e85715...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

76 FITS frames (50 MB), HATP-36250331034216.FITS → HATP-36250331072717.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250331.log (88e963115266...), FinalLightCurve_HAT-P-36 b_2025-03-30.png (bd26e73d70a3...), FinalLightCurve_HAT-P-36 b_2025-03-30.csv (1736d4ae75b2...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 10:32Z.